SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1710**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 30%

Duration : 45 Minutes

QUESTIONS: 2 **Total Marks:20**

Annexure A

Descriptive Written Test

Code of Conduct:

I K.Sasidhar Reddy(192525140) *certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:*K.Sasidhar Reddy*

Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.

2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

IoT - AT3

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(1) A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preference over time. In this context, explain the different types of intelligent agent and analyze how each type would function within this system.

(A) Intelligent agent in a smart home:

Simple reflex:- uses condition action rules on the current percept only.

Ex:- if motion detected and it is night, turn on the light. It is fast and cheap but has no memory, so it can't tell whether a room is empty or the user is just sitting still

Model based reflex:- keeps an internal state of home.

Ex:- if refers the user still in the living room despite no motion for 5 minutes, so it keeps the light on

Goal based reflex:- choose action to each a goal such room at 24°c by 7PM or house secured by 11PM. It plans ahead for example precooling the house before return.

utility - based:- maximizes a utility function that bal comfort, energy cost and security.

ex: when comfort and electricity cost conflict, it picks the AC setting with the best overall score, It also copes with uncertainity.

Best choice:- a hybrid, with a learning component offers rules for immediate safety and security, where speed matter

* A model based layer for tract the home's state
* Goal based planning for scheduled routines
* A utility-based layer for trade-offs between comfort and energy.
* A ~~uti~~ learning element that updates user preferences over time.

This combination gives fast reaction context awareness and efficiency without sacrifing comfort.

2) A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uniformed search strategies like uniform cost search and Iterative Deepening search (IDS) can help solve this problem. Discuss the advantages and disadvantages of those algorithms in items of intelligent agent and explain how agent design effect decision making.

Uniform cost search (UCS): Expands the nodes with the lowest Path cost ($g(n)$) using a priority queues. In maze with varying steps costs; it finds the cheapest Path.

Iterative Deepening search (IDS):

Run depth-limited DFS with limits 0, 1, 2 and soon. It is complete, optimal for uniform step costs and uses little memory. It suits an unknown maze because the goal depth is not known.

Uses Pro and Cons:- Handle variable costs, but it memory use is exponential and it explores in all directions.

IDPS pros and cons: very memory-efficient, but it re-expands upper levels and ignores step costs.

Link to agent types:- A simple reflex agent can loop forever in a maze

* A model based agent remembers visited cells and build a map.
* A goal based agent uses search to plan a path to exist.
* A utility based agent chooses among paths using time, energy and risk.

Agent design decides what the robot remembers and how it chooses between actions, so the search strategy only works well if the agent has a model of the maze.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	**Max Marks**
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand